# A COMMUTATIVE SQUARE OF SITES WITH COCONTINUOUS FUNCTORS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The auxiliary categories of morphisms into the image. Same chapter, Section section-more-functoriality-PSh, where the auxiliary categories are defined. Supplies the categories indexing the limits computed in the proof.
- Sheafification is unnecessary for simultaneously continuous and cocontinuous functors. Same chapter, Lemma lemma-when-shriek. Identifies the pullbacks with the functors on presheaves.
- Initial objects and limits. Categories, Lemma categories-lemma-initial. Shows that the two limits indexed by the two auxiliary categories agree.

*Source reference.* Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 6880–6880. Navigation pointer invoked at proof line 6880; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original section 00XF: sources/stacks-sites/pinned-original-source.tex, source lines 6894–6894. Navigation pointer invoked at proof line 6894; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

---

*Date*: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00XR: sources/stacks-sites/pinned-original-source.tex, source lines 6857–6857. Navigation pointer invoked at proof line 6857; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 002R: sources/stacks-sites/pinned-original-source.tex, source lines 6911–6911. Navigation pointer invoked at proof line 6911; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Assume given sites $\mathcal{C}', \mathcal{C}, \mathcal{D}', \mathcal{D}$ and functors*

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow{v'} & \mathcal{C} \\ {\scriptstyle u'}\downarrow & & \downarrow{\scriptstyle u} \\ \mathcal{D}' & \xrightarrow{v} & \mathcal{D} \end{array}$$

*Assume*

(1) *$u$, $u'$, $v$, and $v'$ are cocontinuous giving rise to morphisms of topoi $f$, $f'$, $g$, and $g'$ by Lemma 00XO,*
(2) *$v \circ u' = u \circ v'$,*
(3) *$v$ and $v'$ are continuous as well as cocontinuous, and*
(4) *for any object $V'$ of $\mathcal{D}'$ the functor ${}^{u'}_{V'}\mathcal{I} \to {}^{u}_{v(V')}\mathcal{I}$ given by $v$ is cofinal.*

*Then $f'_* \circ (g')^{-1} = g^{-1} \circ f_*$ and $g'_! \circ (f')^{-1} = f^{-1} \circ g_!$.*

*Proof.* The categories ${}^{u'}_{V'}\mathcal{I}$ and ${}^{u}_{v(V')}\mathcal{I}$ are defined in Section 00XF. The functor in condition (4) sends the object $\psi : u'(U') \to V'$ of ${}^{u'}_{V'}\mathcal{I}$ to the object $v(\psi) : uv'(U') = vu'(U') \to v(V')$ of ${}^{u}_{v(V')}\mathcal{I}$. Recall that $g^{-1}$ is given by $v^p$ (Lemma 00XR) and $f_*$ is given by ${}_s u = {}_p u$. Hence the value of $g^{-1}f_*\mathcal{F}$ on $V'$ is the value of ${}_p u\mathcal{F}$ on $v(V')$ which is the limit

$$\lim_{u(U) \to v(V') \in \mathrm{Ob}({}^{u}_{v(V')}\mathcal{I}^{opp})} \mathcal{F}(U)$$

By the same reasoning, the value of $f'_*(g')^{-1}\mathcal{F}$ on $V'$ is given by the limit

$$\lim_{u'(U') \to V' \in \mathrm{Ob}({}^{u'}_{V'}\mathcal{I}^{opp})} \mathcal{F}(v'(U'))$$

Thus assumption (4) and Categories, Lemma 002R show that these agree and the first equality of the lemma is proved. The second equality follows from the first by uniqueness of adjoints. □